A portable `std::simd` FFT, up to 3.4× faster than GNU Radio 4's

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Abstract

GNU Radio 4 computes FFTs with `SimdFFT`, a `std::simd` port of PFFFT whose vectors are fixed at four lanes. We present `fastfft`, a single-header radix-8/16 Stockham FFT written against `native_simd` without ISA intrinsics. On one AVX-512 core it is 1.6–3.4× faster than `SimdFFT` for complex single-precision transforms of size 2^6–2^{20} and 1.3–2.0× in double precision, and between 0.86–2.04× (float) and 0.70–1.20× (double) as fast as FFTW 3.3.10 with AVX-512 codelets. Inside GNU Radio 4, a small patch makes the streaming FFT block 1.7–2.5× faster (float). C++26 `std::simd` in GCC trunk removes the last compiler extension. The code and a harness that repeats all measurements on any Linux host are public.

1 Introduction

GNU Radio 4 (gr4) [1] is a header-only C++23 rewrite of the GNU Radio signal-processing framework. Its FFT blocks call `gr::algorithm::FFT`, which dispatches to `SimdFFT`, a port of PFFFT [2] to `std::experimental::simd` [3] via vir-simd [5]. `SimdFFT` uses `simd<T, 4>`: four floats or four doubles, whatever the hardware offers. On an AVX-512 machine this leaves three quarters of each register unused in single precision.

We asked how fast a power-of-two complex FFT can be when written only against `native_simd`, so that the width follows the compilation target, and whether it can match a tuned library. The answer, `fastfft`, is one 400 line header.

2 Algorithm

`fastfft` is a Stockham autosort FFT [7, 8]: every pass reads one buffer and writes the other in natural order, so no bit-reversal pass is needed. Data are interleaved complex numbers; a vector of W scalars holds $C = W/2$ complex values (8 floats or 4 doubles per 512 bit register).

1. *First pass, radix 8, vectorised over the butterfly index.* With stride 1 the eight inputs of a butterfly are $N/8$ apart, so C consecutive butterflies form one vector per input. The eight results of each butterfly must be stored contiguously; an in-register $8 \times C$ complex transpose (three rounds of two-input zips) does this before the store.
2. *Middle passes, radix 16, vectorised over the stride.* Once the stride $s \geq C$, the C lanes of a vector are C independent sub-transforms that share a twiddle factor, which is broadcast. A radix-16 butterfly is done as 4×4 in registers. Each pass reads and writes the data once, so a 2^{20} transform takes six passes instead of ten for radix 4.
3. *Last pass,* radix 2, 4 or 8, whatever remains.

Complex multiplication by a twiddle w uses tables stored as (w_r, w_r) and $(-w_i, w_i)$ per lane pair, so $z \cdot w = z \odot \mathbf{w_r} + \mathrm{swap}(z) \odot \mathbf{w'_i}$ costs one multiply, one fused multiply-add and one in-lane swap. Twiddles are precomputed in double precision. Sizes below 64 use a scalar radix-2 path.

3 What `std::simd` could not express

The kernel uses `std::experimental::simd` as shipped with libstdc++ and contains no intrinsics, but three points needed care. **(i) Two-input shuffles.** The transpose in pass 1 needs zip operations that take lanes from two vectors. `std::simd` has none; `concat` followed by a permute compiled to element-wise moves that cost about three times the whole pass. We use the compiler's generic vector extension (`__builtin_shufflevector`), which GCC and clang lower to permute instructions (`vpermt2ps` on AVX-512). The re/im swap is written the same way. **(ii) FMA.** libstdc++'s `fma(simd,simd,simd)` is not inlined; we write `a*b+c` and compile with `-ffp-contract=fast` (GCC's default in GNU mode). **(iii) Backends.** vir-simd forwards to libstdc++'s implementation when it is available; with libc++ it falls back to emulating the vector types with arrays, silently (§5). Our header defines `FASTFFT_NATIVE_SIMD` so that callers can detect this case. **(iv) C++26.** GCC trunk (17.0.0 20261010) ships C++26 `std::simd` [4]. There (i) has a standard spelling, `permute<W>(cat(a,b), f)`, which GCC lowers to the same `vpermt2ps`; with `-std=c++26` the header uses it and needs no compiler extension. It also offers `vec<complex<T>>`, which we tried as well (§5).

4 Method

`fftbench` times `fastfft`, `SimdFFT` (gr4 rev. `af62235`) and FFTW 3.3.10 [6] (built from source with the host's SIMD codelets, `FFTW_MEASURE`) on the same input, uniform random in $[-1, 1)^2$. A *sample* transforms $\max(N, 2^{16})$ consecutive values as frames of N, the way a streaming gr4 block sees them, repeated until it lasts at least 5 ms; one warm-up sample is discarded. Before timing, every output is checked against a long-double reference DFT. Each process runs 40 rounds; in every round each implementation is sampled once in a freshly randomised order, so drifts affect all of them alike and same-round samples can be paired. The process is run three times and pinned to one core. We report the median of the 120 samples per cell with a 95 % percentile-bootstrap confidence interval, and speedups as the median of paired per-round ratios, also with a bootstrap CI [9, 10]. All values, raw samples and host data are in the repository; one script (`bench/run.sh`) repeats the whole procedure.

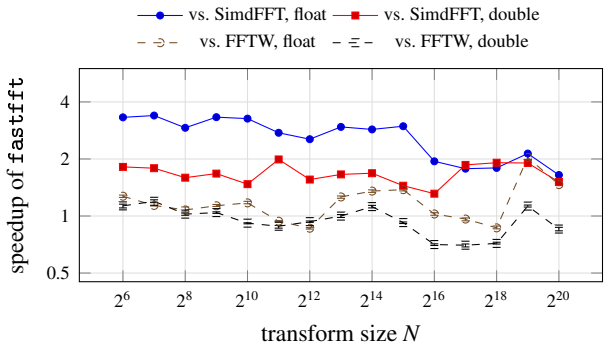


Figure 1: Speedup of `fastfft` (median of paired ratios, 95 % CI as error bars; mostly smaller than the markers). Above 1 means `fastfft` is faster. Absolute times in Tables 1 and 2.

Host: virtual machine with 4 vCPU of an Intel Xeon at 2.1 GHz nominal (AVX-512 incl. FP16, 2 MiB L2 per core), Linux 6.18.44-fc-v114, g++-14 14.2.0, `-O3 -march=native`. The largest relative median absolute deviation over all cells was 10.2 %.

5 Results

Kernel. Figure 1 summarises the comparison. In single precision `fastfft` is 2.5–3.4× faster than `SimdFFT` up to 2^{15} and 1.6–2.1× above, where both are increasingly bound by memory traffic. Against FFTW the picture depends on N (0.86–2.04×): `fastfft` is ahead at 2^6–2^{10}, 2^{13}–2^{15} and 2^{19}–2^{20}, within 5 % at 2^{16}–2^{17}, and behind at 2^{11}–2^{12} and 2^{18}. In double precision the gain over `SimdFFT` is 1.3–2.0×, smaller because four doubles already fill half an AVX-512 register; `fastfft` beats FFTW up to 2^9 and at 2^{14} and 2^{19}, is level at 2^{13}, and FFTW is faster at 2^{10}–2^{12}, 2^{15}–2^{18} (down to 0.70×) and 2^{20}. Accuracy is unaffected: the largest error relative to $\max|X|$ was 2.1×10^{-7} (float) and 6.0×10^{-16} (double), against 4.8×10^{-7} and 8.4×10^{-16} for `SimdFFT` (Table 7).

Vector width. Compiling `fastfft` for narrower vectors separates the effect of width from that of the algorithm (Table 3). With 128 bit vectors, the width `SimdFFT` uses for floats, `fastfft` is 0.9–1.8× as fast as `SimdFFT`: the larger radices help mainly for small N. Going from 128 bit to 512 bit vectors then adds another factor of 1.4–3.0.

C++26 `std::simd`. Built with GCC trunk, the extension-free C++26 path is 1.05–1.25× slower than the same compiler's `<experimental/simd>` path for $N \le 2^9$ (double 1.09–1.17×) and mostly level above, apart from 2^{20} (Tables 5, 6; separate processes, so not paired). Against `SimdFFT` built with the same compiler it remains 1.6–2.9× faster (double 1.4–2.0×). We also rewrote the kernel on `std::simd::vec<std::complex<T>>` (`fastfft::cx`, `include/fastfft/fft_cx.hpp`). Products and $\pm i$ rotations become plain complex arithmetic, and the twiddle tables halve. Three things in the current implementation got in the way: `permute` does not accept complex vectors, so the zips go through `real()`/`imag()` and back; broadcasting a complex scalar compiles to scalar stores, so we build it from two real broadcasts; and `operator*` carries the Annex G NaN/Inf recovery unless compiled with `-fcx-limited-range`. Accuracy is unchanged (2.0×10^{-7}). `fastfft::cx` is 1.13–1.48× slower up to 2^{15} but takes 0.88–0.95× the time of the real-lane version from 2^{16} (double: 0.88–1.27 and 0.88–0.92), where the smaller twiddle tables likely help. Replacing the zip and broadcast workarounds by a non-portable `bit_cast` to a real vector reduces the small-size gap to 0.99–1.18×; the remainder is the generic complex product (three in-lane shuffles where our pre-arranged twiddles need one).

Emulated `simd`. Built with clang 20 and libc++, where vir-simd emulates the vector types, `fastfft` becomes 3.6–11.3× slower and `SimdFFT` 1.2–2.0× slower (Table 4). Under emulation `fastfft` is *slower* than `SimdFFT` for $N \ge 2^7$, as its single-instruction shuffles become element loops, so the patch below enables it only with native types.

In gr4. The patch (`gr4/0001-fastfft.patch`) adds the header to gr4 and makes `gr::algorithm::FFT` use it for complex power-of-two sizes ≥ 64; real-input and non-power-of-two transforms still use `SimdFFT`. The spectrum block also reuses its output buffer and fuses magnitude, phase and fftshift into one pass. gr4's FFT unit tests pass unchanged (GCC 14.2). Measured with gr4's own benchmark facility (Table 8, 5 alternating runs per build), the streaming block `FFT<T>` gets 1.7–2.5× faster for floats and 1.3–1.8× for doubles. The spectrum block `FFT<T,DataSet<P>>` (`bm_fft_block`, $N = 1024$) is unchanged within noise for complex float and real input (1.06×, 1.00×); the 1.55× for complex double we have not analysed. For complex float input it spends about 70 µs per frame, of which the FFT takes 0.7 µs; the rest goes to `hypot`, `atan2` and building the `DataSet`.

6 Discussion

Using the native vector width and fewer, larger radix passes is enough to make a portable FFT competitive with a hand-tuned library on the measured host; nothing in the kernel is specific to x86. FFTW's remaining lead at some sizes suggests that choosing radices per size, as FFTW's planner does, would help further. With C++26 `std::simd` the kernel is pure standard C++, at a cost of 5–25 % for small N on today's trunk; `vec<complex<T>>` reads most naturally and is already faster for large N, and would be for all N once `permute`, broadcast and the complex product are optimised for it. The emulation result shows the other side of portability: code written for native vector types can become slower than simpler code when the types are emulated, so a library should check which backend it got.

These results come from one virtual AVX-512 core; other CPUs and compilers may differ, and collecting their data takes one command (`bench/run.sh`).

Code and contributions. `https://github.com/dennisklein/fastfft` (headers, harness, gr4 patch, raw data). D.K. posed the question, steered the work and reviewed the results. Claude, an AI model by Anthropic, wrote the code, ran the measurements and drafted the text.

References

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[3] ISO/IEC TS 19570:2018, Parallelism TS v2 (`std::experimental::simd`).

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Appendix: measurement data

Table 1: Complex float: median ns per transform [95 % CI] and speedup of fastfft (median of paired ratios [95 % CI]).

N	fastfft	gr4 SimdFFT	FFTW 3.3.10	vs. SimdFFT	vs. FFTW
2^6	35.8 [34.8, 36.7]	117 [116, 122]	45.6 [44.9, 46.5]	3.32 [3.28, 3.37]	1.27 [1.25, 1.29]
2^7	69.8 [69.5, 70.2]	236 [235, 239]	79.6 [78.8, 84.7]	3.39 [3.38, 3.42]	1.14 [1.13, 1.18]
2^8	163 [161, 165]	471 [469, 475]	180 [172, 182]	2.92 [2.89, 2.93]	1.08 [1.05, 1.12]
2^9	315 [311, 318]	1035 [1028, 1051]	361 [353, 365]	3.32 [3.29, 3.36]	1.13 [1.12, 1.15]
2^{10}	658 [650, 663]	2123 [2113, 2142]	769 [755, 804]	3.27 [3.25, 3.30]	1.17 [1.16, 1.20]
2^{11}	1800 [1789, 1813]	4967 [4947, 4999]	1709 [1689, 1723]	2.75 [2.74, 2.76]	0.94 [0.93, 0.95]
2^{12}	4926 [4906, 4952]	12.5 k [12.3 k, 12.9 k]	4245 [4222, 4284]	2.54 [2.52, 2.58]	0.86 [0.85, 0.87]
2^{13}	10.3 k [10.2 k, 10.4 k]	30.1 k [29.9 k, 30.6 k]	12.9 k [12.8 k, 13.1 k]	2.95 [2.93, 2.97]	1.26 [1.24, 1.28]
2^{14}	21.4 k [21.1 k, 21.6 k]	60.9 k [60.4 k, 61.4 k]	28.6 k [28.3 k, 29.1 k]	2.86 [2.84, 2.89]	1.35 [1.33, 1.37]
2^{15}	45.0 k [44.5 k, 45.6 k]	133.0 k [132.7 k, 133.6 k]	61.9 k [61.5 k, 62.6 k]	2.98 [2.96, 3.00]	1.38 [1.36, 1.39]
2^{16}	143.2 k [141.3 k, 145.7 k]	275.8 k [274.0 k, 278.1 k]	146.7 k [141.1 k, 150.2 k]	1.94 [1.92, 1.96]	1.02 [1.01, 1.04]
2^{17}	411.5 k [406.9 k, 416.6 k]	734.3 k [721.2 k, 742.5 k]	394.9 k [389.4 k, 400.1 k]	1.78 [1.76, 1.80]	0.96 [0.94, 0.98]
2^{18}	1.031 M [1.021 M, 1.042 M]	1.835 M [1.814 M, 1.851 M]	898.7 k [886.1 k, 911.2 k]	1.79 [1.78, 1.81]	0.87 [0.86, 0.89]
2^{19}	2.105 M [2.083 M, 2.121 M]	4.429 M [4.378 M, 4.506 M]	4.373 M [4.293 M, 4.423 M]	2.14 [2.11, 2.15]	2.04 [1.99, 2.09]
2^{20}	5.474 M [5.427 M, 5.542 M]	8.954 M [8.908 M, 9.016 M]	8.029 M [7.976 M, 8.078 M]	1.65 [1.64, 1.66]	1.46 [1.46, 1.49]

Table 2: Complex double: median ns per transform [95 % CI] and speedup of fastfft (median of paired ratios [95 % CI]).

N	fastfft	gr4 SimdFFT	FFTW 3.3.10	vs. SimdFFT	vs. FFTW
2^6	74.7 [73.2, 76.2]	132 [131, 135]	84.6 [83.2, 85.4]	1.82 [1.80, 1.83]	1.13 [1.10, 1.15]
2^7	161 [159, 164]	286 [284, 288]	193 [191, 194]	1.79 [1.77, 1.80]	1.20 [1.18, 1.22]
2^8	382 [379, 385]	605 [601, 610]	396 [391, 401]	1.59 [1.58, 1.61]	1.02 [1.01, 1.04]
2^9	837 [827, 851]	1389 [1377, 1409]	884 [873, 902]	1.67 [1.66, 1.69]	1.04 [1.03, 1.05]
2^{10}	2146 [2130, 2163]	3165 [3144, 3181]	1958 [1939, 1975]	1.47 [1.47, 1.48]	0.92 [0.91, 0.92]
2^{11}	4992 [4953, 5017]	9923 [9854, 9970]	4389 [4333, 4520]	1.99 [1.98, 2.01]	0.88 [0.86, 0.89]
2^{12}	11.8 k [11.7 k, 12.0 k]	18.2 k [18.1 k, 18.4 k]	10.9 k [10.8 k, 11.3 k]	1.56 [1.54, 1.58]	0.93 [0.93, 0.94]
2^{13}	24.7 k [24.4 k, 25.0 k]	40.7 k [40.5 k, 41.0 k]	24.7 k [24.4 k, 24.9 k]	1.66 [1.65, 1.68]	1.00 [0.98, 1.02]
2^{14}	52.4 k [52.0 k, 52.9 k]	88.5 k [87.7 k, 89.2 k]	58.2 k [57.3 k, 58.8 k]	1.68 [1.67, 1.69]	1.12 [1.10, 1.14]
2^{15}	136.0 k [134.0 k, 143.3 k]	198.7 k [196.9 k, 204.6 k]	125.6 k [123.8 k, 130.5 k]	1.45 [1.42, 1.46]	0.92 [0.91, 0.93]
2^{16}	405.7 k [400.0 k, 409.7 k]	523.1 k [519.1 k, 528.0 k]	284.3 k [281.4 k, 287.0 k]	1.31 [1.29, 1.33]	0.71 [0.70, 0.71]
2^{17}	999.9 k [991.1 k, 1.008 M]	1.866 M [1.850 M, 1.876 M]	702.1 k [687.6 k, 712.8 k]	1.86 [1.84, 1.87]	0.70 [0.69, 0.71]
2^{18}	2.083 M [2.059 M, 2.100 M]	4.027 M [3.977 M, 4.054 M]	1.486 M [1.471 M, 1.495 M]	1.91 [1.90, 1.94]	0.72 [0.71, 0.73]
2^{19}	4.599 M [4.536 M, 4.679 M]	8.812 M [8.739 M, 8.854 M]	5.187 M [5.157 M, 5.221 M]	1.90 [1.89, 1.93]	1.13 [1.11, 1.14]
2^{20}	12.501 M [12.066 M, 12.788 M]	18.515 M [18.254 M, 18.966 M]	10.232 M [10.180 M, 10.375 M]	1.51 [1.45, 1.57]	0.85 [0.84, 0.87]

Table 3: Vector width, complex float: fastfft compiled for 128, 256, 512-bit vectors (`FASTFFT_SIMD_BYTES`), median ns per transform [95 % CI]; SimdFFT for reference.

N	128 bit	256 bit	512 bit	SimdFFT
2^6	65.2 [64.7, 65.9]	48.2 [48.0, 48.4]	35.8 [34.8, 36.7]	117 [116, 122]
2^7	134 [133, 134]	101 [101, 101]	69.8 [69.5, 70.2]	236 [235, 239]
2^8	337 [334, 342]	222 [220, 223]	163 [161, 165]	471 [469, 475]
2^9	682 [678, 688]	448 [446, 451]	315 [311, 318]	1035 [1028, 1051]
2^{10}	1490 [1483, 1502]	989 [978, 1004]	658 [650, 663]	2123 [2113, 2142]
2^{11}	3300 [3280, 3316]	2272 [2260, 2311]	1800 [1789, 1813]	4967 [4947, 4999]
2^{12}	9133 [8964, 9289]	6146 [6123, 6175]	4926 [4906, 4952]	12.5 k [12.3 k, 12.9 k]
2^{13}	18.9 k [18.7 k, 19.0 k]	13.2 k [13.1 k, 13.3 k]	10.3 k [10.2 k, 10.4 k]	30.1 k [29.9 k, 30.6 k]
2^{14}	41.5 k [39.8 k, 42.2 k]	27.6 k [27.5 k, 27.9 k]	21.4 k [21.1 k, 21.6 k]	60.9 k [60.4 k, 61.4 k]
2^{15}	135.7 k [133.8 k, 140.4 k]	74.6 k [74.0 k, 75.3 k]	45.0 k [44.5 k, 45.6 k]	133.0 k [132.7 k, 133.6 k]
2^{16}	299.8 k [297.0 k, 305.3 k]	184.0 k [183.1 k, 185.3 k]	143.2 k [141.3 k, 145.7 k]	275.8 k [274.0 k, 278.1 k]
2^{17}	641.1 k [639.7 k, 644.8 k]	449.9 k [447.1 k, 453.7 k]	411.5 k [406.9 k, 416.6 k]	734.3 k [721.2 k, 742.5 k]
2^{18}	1.413 M [1.407 M, 1.417 M]	1.070 M [1.066 M, 1.076 M]	1.031 M [1.021 M, 1.042 M]	1.835 M [1.814 M, 1.851 M]
2^{19}	3.826 M [3.769 M, 3.896 M]	2.541 M [2.528 M, 2.552 M]	2.105 M [2.083 M, 2.121 M]	4.429 M [4.378 M, 4.506 M]
2^{20}	8.354 M [8.325 M, 8.396 M]	6.035 M [6.011 M, 6.068 M]	5.474 M [5.427 M, 5.542 M]	8.954 M [8.908 M, 9.016 M]

Table 4: Standard library: complex float built with GCC/libstdc++ (native simd) and with clang/libc++ (vir-simd array emulation), median ns per transform [95 % CI].

N	fastfft, libstdc++	fastfft, libc++	SimdFFT, libstdc++	SimdFFT, libc++
2^6	35.8 [34.8, 36.7]	165 [162, 167]	117 [116, 122]	232 [230, 236]
2^7	69.8 [69.5, 70.2]	791 [784, 799]	236 [235, 239]	456 [454, 460]
2^8	163 [161, 165]	1493 [1483, 1524]	471 [469, 475]	889 [882, 903]
2^9	315 [311, 318]	2881 [2843, 2909]	1035 [1028, 1051]	1990 [1975, 2000]
2^{10}	658 [650, 663]	6423 [6379, 6482]	2123 [2113, 2142]	4143 [4119, 4167]
2^{11}	1800 [1789, 1813]	18.2 k [18.1 k, 18.3 k]	4967 [4947, 4999]	9284 [9208, 9353]
2^{12}	4926 [4906, 4952]	37.0 k [36.7 k, 37.1 k]	12.5 k [12.3 k, 12.9 k]	21.0 k [20.9 k, 21.2 k]
2^{13}	10.3 k [10.2 k, 10.4 k]	74.6 k [73.7 k, 76.5 k]	30.1 k [29.9 k, 30.6 k]	46.8 k [46.4 k, 48.1 k]
2^{14}	21.4 k [21.1 k, 21.6 k]	160.7 k [159.8 k, 165.5 k]	60.9 k [60.4 k, 61.4 k]	98.8 k [97.2 k, 105.4 k]
2^{15}	45.0 k [44.5 k, 45.6 k]	401.9 k [399.3 k, 405.2 k]	133.0 k [132.7 k, 133.6 k]	209.7 k [208.0 k, 212.4 k]
2^{16}	143.2 k [141.3 k, 145.7 k]	839.8 k [833.3 k, 856.2 k]	275.8 k [274.0 k, 278.1 k]	443.7 k [438.0 k, 451.4 k]
2^{17}	411.5 k [406.9 k, 416.6 k]	1.756 M [1.736 M, 1.779 M]	734.3 k [721.2 k, 742.5 k]	1.038 M [1.029 M, 1.057 M]
2^{18}	1.031 M [1.021 M, 1.042 M]	3.792 M [3.758 M, 3.837 M]	1.835 M [1.814 M, 1.851 M]	2.379 M [2.359 M, 2.410 M]
2^{19}	2.105 M [2.083 M, 2.121 M]	9.006 M [8.951 M, 9.051 M]	4.429 M [4.378 M, 4.506 M]	5.314 M [5.237 M, 5.347 M]
2^{20}	5.474 M [5.427 M, 5.542 M]	19.933 M [19.450 M, 20.493 M]	8.954 M [8.908 M, 9.016 M]	11.341 M [11.250 M, 11.429 M]

Table 5: GCC trunk (17.0) with `-std=c++26`, complex float: median ns per transform [95 % CI]. fastfft on `<experimental/simd>` (GCC 14 and trunk), on C++26 `std::simd`, and fastfft::cx on `std::simd::vec<std::complex<float>>` (plain, and with zip/broadcast via `bit_cast` to a real vector); SimdFFT built with trunk for reference.

N	exp., GCC 14	exp., trunk	C++26	cx	cx, bit_cast	SimdFFT
2^6	35.8	35.3	44.0	57.8	46.7	106
	[34.8, 36.7]	[35.1, 35.5]	[43.7, 44.6]	[57.1, 58.5]	[46.4, 46.9]	[105, 107]
2^7	69.8	75.7	83.8	108	92.6	219
	[69.5, 70.2]	[72.1, 86.3]	[83.5, 84.4]	[108, 109]	[92.1, 93.3]	[218, 220]
2^8	163	173	182	240	198	445
	[161, 165]	[167, 176]	[181, 184]	[239, 242]	[197, 200]	[442, 447]
2^9	315	302	340	499	389	988
	[311, 318]	[299, 307]	[336, 343]	[496, 507]	[387, 392]	[981, 997]
2^{10}	658	654	719	1063	817	2096
	[650, 663]	[641, 660]	[703, 742]	[1053, 1079]	[813, 821]	[2072, 2163]
2^{11}	1800	1813	1809	2257	1931	4809
	[1789, 1813]	[1799, 1824]	[1805, 1819]	[2245, 2271]	[1907, 1952]	[4782, 4853]
2^{12}	4926	5043	5020	5814	4970	12.1 k
	[4906, 4952]	[5000, 5089]	[4996, 5056]	[5777, 5858]	[4932, 4998]	[12.1 k, 12.2 k]
2^{13}	10.3 k	10.2 k	10.7 k	12.6 k	10.7 k	29.3 k
	[10.2 k, 10.4 k]	[10.0 k, 10.3 k]	[10.5 k, 10.8 k]	[12.6 k, 12.7 k]	[10.6 k, 11.1 k]	[29.2 k, 29.4 k]
2^{14}	21.4 k	20.5 k	21.4 k	25.9 k	20.9 k	59.0 k
	[21.1 k, 21.6 k]	[20.4 k, 20.6 k]	[21.1 k, 21.7 k]	[25.7 k, 26.3 k]	[20.8 k, 21.1 k]	[58.6 k, 59.4 k]
2^{15}	45.0 k	44.2 k	47.7 k	54.1 k	46.0 k	134.2 k
	[44.5 k, 45.6 k]	[44.0 k, 44.5 k]	[47.4 k, 48.1 k]	[53.8 k, 54.8 k]	[45.8 k, 46.5 k]	[133.5 k, 135.0 k]
2^{16}	143.2 k	139.5 k	136.9 k	129.4 k	120.6 k	273.4 k
	[141.3 k, 145.7 k]	[137.8 k, 143.9 k]	[135.6 k, 137.9 k]	[128.4 k, 130.4 k]	[119.1 k, 122.0 k]	[272.4 k, 275.1 k]
2^{17}	411.5 k	395.9 k	398.9 k	353.7 k	337.5 k	717.0 k
	[406.9 k, 416.6 k]	[390.5 k, 398.2 k]	[388.3 k, 403.1 k]	[349.6 k, 358.1 k]	[335.3 k, 340.0 k]	[706.2 k, 730.2 k]
2^{18}	1.031 M	1.009 M	1.005 M	883.7 k	908.8 k	1.805 M
	[1.021 M, 1.042 M]	[997.7 k, 1.019 M]	[998.9 k, 1.013 M]	[880.2 k, 890.0 k]	[901.5 k, 915.2 k]	[1.794 M, 1.812 M]
2^{19}	2.105 M	2.041 M	2.147 M	1.902 M	1.948 M	4.531 M
	[2.083 M, 2.121 M]	[2.034 M, 2.053 M]	[2.120 M, 2.173 M]	[1.887 M, 1.920 M]	[1.927 M, 1.960 M]	[4.470 M, 4.587 M]
2^{20}	5.474 M	4.957 M	5.837 M	5.116 M	4.680 M	9.211 M
	[5.427 M, 5.542 M]	[4.865 M, 5.049 M]	[5.736 M, 5.979 M]	[5.068 M, 5.231 M]	[4.654 M, 4.705 M]	[9.058 M, 9.323 M]

Table 6: GCC trunk (17.0) with `-std=c++26`, complex double: median ns per transform [95 % CI]. fastfft on `<experimental/simd>` (GCC 14 and trunk), on C++26 `std::simd`, and fastfft::cx on `std::simd::vec<std::complex<double>>` (plain, and with zip/broadcast via `bit_cast` to a real vector); SimdFFT built with trunk for reference.

N	exp., GCC 14	exp., trunk	C++26	cx	cx, bit_cast	SimdFFT
2^6	74.7	78.6	87.1	108	94.3	132
	[73.2, 76.2]	[75.7, 79.7]	[85.7, 88.3]	[107, 110]	[91.8, 98.8]	[131, 133]
2^7	161	164	179	227	191	284
	[159, 164]	[162, 167]	[176, 181]	[226, 229]	[189, 193]	[281, 286]
2^8	382	372	434	483	455	608
	[379, 385]	[368, 374]	[430, 440]	[479, 488]	[451, 459]	[602, 610]
2^9	837	813	944	1092	954	1444
	[827, 851]	[807, 822]	[934, 954]	[1078, 1107]	[940, 967]	[1414, 1485]
2^{10}	2146	2185	2206	2282	2114	3251
	[2130, 2163]	[2166, 2209]	[2195, 2226]	[2274, 2310]	[2099, 2129]	[3237, 3288]
2^{11}	4992	5105	4962	4984	4939	10.1 k
	[4953, 5017]	[5047, 5165]	[4907, 5009]	[4948, 5030]	[4870, 4994]	[10.0 k, 10.1 k]
2^{12}	11.8 k	11.5 k	12.0 k	12.7 k	11.8 k	18.0 k
	[11.7 k, 12.0 k]	[11.4 k, 11.6 k]	[11.9 k, 12.0 k]	[12.6 k, 12.8 k]	[11.6 k, 12.2 k]	[18.0 k, 18.1 k]
2^{13}	24.7 k	23.8 k	24.4 k	26.1 k	23.6 k	39.9 k
	[24.4 k, 25.0 k]	[23.6 k, 24.0 k]	[24.3 k, 24.6 k]	[25.8 k, 26.2 k]	[23.4 k, 23.9 k]	[39.8 k, 40.1 k]
2^{14}	52.4 k	49.6 k	52.9 k	54.1 k	49.6 k	87.6 k
	[52.0 k, 52.9 k]	[49.4 k, 49.9 k]	[52.2 k, 53.4 k]	[53.7 k, 54.4 k]	[49.4 k, 49.7 k]	[87.3 k, 88.0 k]
2^{15}	136.0 k	135.9 k	133.4 k	117.0 k	114.4 k	200.5 k
	[134.0 k, 143.3 k]	[133.7 k, 137.7 k]	[132.1 k, 135.2 k]	[116.6 k, 117.9 k]	[112.4 k, 115.2 k]	[199.5 k, 201.4 k]
2^{16}	405.7 k	399.7 k	386.0 k	354.8 k	344.9 k	540.2 k
	[400.0 k, 409.7 k]	[394.2 k, 404.0 k]	[382.6 k, 402.1 k]	[350.5 k, 361.5 k]	[343.0 k, 347.1 k]	[532.0 k, 549.1 k]
2^{17}	999.9 k	1.008 M	1.015 M	899.3 k	880.9 k	1.860 M
	[991.1 k, 1.008 M]	[998.0 k, 1.019 M]	[1.005 M, 1.027 M]	[891.9 k, 905.9 k]	[874.2 k, 885.1 k]	[1.836 M, 1.881 M]
2^{18}	2.083 M	2.103 M	2.182 M	2.005 M	1.928 M	4.134 M
	[2.059 M, 2.100 M]	[2.070 M, 2.139 M]	[2.149 M, 2.200 M]	[1.983 M, 2.027 M]	[1.912 M, 1.947 M]	[4.089 M, 4.175 M]
2^{19}	4.599 M	4.199 M	5.022 M	4.422 M	3.824 M	8.947 M
	[4.536 M, 4.679 M]	[4.173 M, 4.236 M]	[4.920 M, 5.213 M]	[4.350 M, 4.528 M]	[3.791 M, 3.861 M]	[8.887 M, 9.026 M]
2^{20}	12.501 M	10.151 M	13.141 M	11.766 M	9.922 M	19.523 M
	[12.066 M, 12.788 M]	[10.061 M, 10.241 M]	[12.723 M, 13.383 M]	[11.528 M, 11.989 M]	[9.834 M, 10.069 M]	[18.817 M, 19.845 M]

Table 7: Accuracy: largest $\max_k |X_k - \hat{X}_k| / \max_k |\hat{X}_k|$ over all N, against a long-double reference.

	fastfft	SimdFFT	FFTW
float	2.1×10^{-7}	4.8×10^{-7}	2.1×10^{-7}
double	6.0×10^{-16}	8.4×10^{-16}	3.3×10^{-16}

Table 8: gnuradio4 blocks before and after the patch, gnuradio4's benchmark facility: median call time over runs, speedup as median of paired per-run ratios [95 % CI].

benchmark row	unpatched	patched	speedup
`std::complex<float> - fft`	70.5 µs	68.9 µs	1.06 [0.98, 1.07]
`std::complex<float> - fftCT`	79.7 µs	80.6 µs	1.00 [0.95, 1.01]
`std::complex<double> - fft`	93.0 µs	60.2 µs	1.55 [1.45, 1.59]
`std::complex<double> - fftCT`	98.0 µs	97.9 µs	0.99 [0.91, 1.21]
`float - fft`	23.3 µs	22.7 µs	1.00 [0.97, 1.07]
`double - fft`	27.4 µs	27.7 µs	1.01 [0.81, 1.10]
`FFT<float32> stream N= 64`	138.3 µs	53.5 µs	2.55 [2.53, 4.34]
`FFT<float32> stream N= 256`	135.3 µs	56.6 µs	2.39 [2.33, 4.00]
`FFT<float32> stream N= 1024`	150.4 µs	59.0 µs	2.52 [2.49, 2.57]
`FFT<float32> stream N= 4096`	210.3 µs	97.6 µs	2.14 [2.08, 2.18]
`FFT<float32> stream N= 16384`	256.6 µs	103.9 µs	2.48 [2.37, 2.60]
`FFT<float32> stream N= 65536`	300.1 µs	178.0 µs	1.70 [1.58, 1.73]
`FFT<float32> stream N= 262144`	1979.0 µs	1159.0 µs	1.76 [1.64, 1.83]
`FFT<float32> stream N=1048576`	9300.0 µs	5582.0 µs	1.66 [1.65, 1.73]
`FFT<float64> stream N= 64`	195.0 µs	144.3 µs	1.34 [1.19, 1.43]
`FFT<float64> stream N= 256`	228.3 µs	161.8 µs	1.42 [1.32, 1.47]
`FFT<float64> stream N= 1024`	274.7 µs	205.5 µs	1.34 [1.16, 1.37]
`FFT<float64> stream N= 4096`	367.3 µs	276.4 µs	1.33 [1.26, 1.41]
`FFT<float64> stream N= 16384`	444.5 µs	315.7 µs	1.43 [1.35, 1.60]
`FFT<float64> stream N= 65536`	616.6 µs	490.2 µs	1.33 [1.16, 1.55]
`FFT<float64> stream N= 262144`	4268.0 µs	2326.0 µs	1.84 [1.74, 1.87]
`FFT<float64> stream N=1048576`	19490.0 µs	11470.0 µs	1.74 [1.66, 1.76]